

J2EE Web Components - ITE-5332-0NA

Lab 4

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Application design

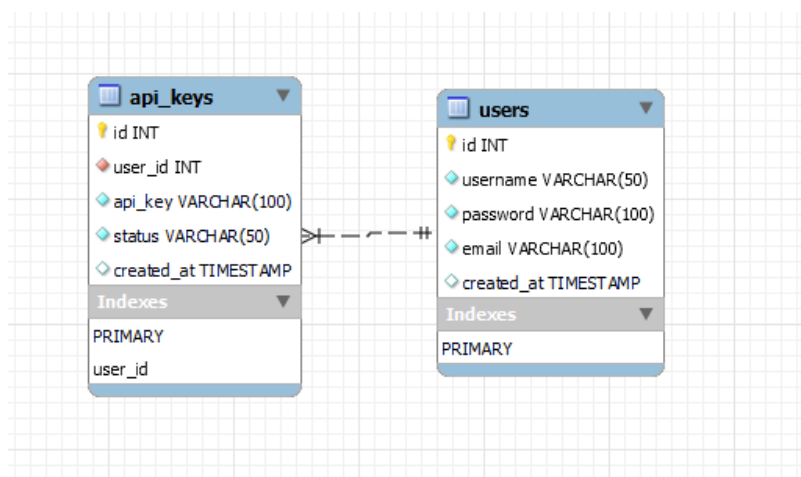
Front-end: html, CSS and JavaScript.

Back-end: Java EE, with business rules layer (Servlets) and data access layer (repository class)

Database: MySQL on AWS

Database schema

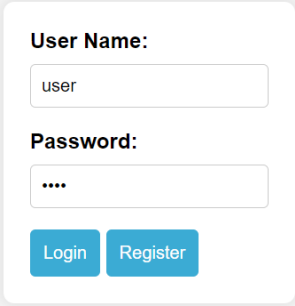
Schema consists of two tables.



Design and Functionalities

Login page

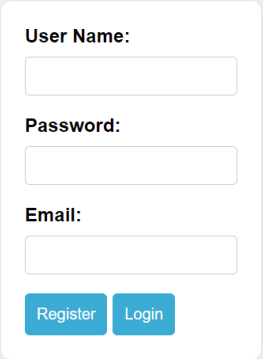
Allow user to input username and password, and login. Or navigate to register page to create new user.



A login form centered on a light gray background. The form is a white rounded rectangle with a subtle shadow. It contains two input fields: 'User Name:' with the text 'user' and 'Password:' with four dots. Below the fields are two blue buttons labeled 'Login' and 'Register'.

Register page

Allow user to input username, password and email to create new user. Or navigate to login page.



A register form centered on a light gray background. The form is a white rounded rectangle with a subtle shadow. It contains three input fields: 'User Name:', 'Password:', and 'Email:'. Below the fields are two blue buttons labeled 'Register' and 'Login'.

Dashboard page

User can view all the keys they have in a table.

Key	Status	Creation Date	Select
bc533ca6-1713-4e06-9993-b68e4d72a179	Active	2024-09-30	☐
22b06a94-20dd-4666-84cd-5aa676930da3	Active	2024-09-30	☐
a5d4f40c-4a63-4ab6-8d4a-4cad4937bacc	Active	2024-09-30	☐
7ec9ab61-29d3-4253-8ac4-203be0b23347	Invalid	2024-09-30	☐
28302d4d-5d98-4c07-8f52-1b61ac956b27	Active	2024-09-30	☐
029522e2-8f89-4705-8784-c1beea7a0989	Active	2024-09-30	☐
db714111-4d98-4b84-bb10-cbc460e72b7b	Active	2024-09-30	☐

New Key

Deactivate

Manage Key

When click the “New Key” button, a new key will be created.

Key	Status	Creation Date	Select
bc533ca6-1713-4e06-9993-b68e4d72a179	Active	2024-09-30	☐
22b06a94-20dd-4666-84cd-5aa676930da3	Active	2024-09-30	☐
a5d4f40c-4a63-4ab6-8d4a-4cad4937bacc	Active	2024-09-30	☐
7ec9ab61-29d3-4253-8ac4-203be0b23347	Invalid	2024-09-30	☐
28302d4d-5d98-4c07-8f52-1b61ac956b27	Active	2024-09-30	☐
029522e2-8f89-4705-8784-c1beea7a0989	Active	2024-09-30	☐
db714111-4d98-4b84-bb10-cbc460e72b7b	Active	2024-09-30	☐
36ea76d1-2fff-44c4-af95-a212a5499fb9	Active	2024-09-30	☐
696d8102-adc1-438c-b99b-4abc08d17027	Active	2024-09-30	☐
3bfa2157-0171-4357-bb3b-0798a84c1ecc	Active	2024-09-30	☐
c49ced9a-59bd-43f5-b0ea-dbf6aa6c7a648	Active	2024-09-30	☐

New Key

Deactivate

Manage Key

When select a key and click “Deactivate” button, the key's status will change.

Key	Status	Creation Date	Select
bc533ca6-1713-4e06-9993-b68e4d72a179	Active	2024-09-30	☐
22b06a94-20dd-4666-84cd-5aa676930da3	Active	2024-09-30	☐
a5d4f40c-4a63-4ab6-8d4a-4cad4937bacc	Active	2024-09-30	☐
7ec9ab61-29d3-4253-8ac4-203be0b23347	Invalid	2024-09-30	☐
28302d4d-5d98-4c07-8f52-1b61ac956b27	Active	2024-09-30	☐
029522e2-8f89-4705-8784-c1beea7a0989	Active	2024-09-30	☐
db714111-4d98-4b84-bb10-cbc460e72b7b	Active	2024-09-30	☐
36ea76d1-2fff-44c4-af95-a212a5499fb9	Active	2024-09-30	☐
696d8102-adc1-438c-b99b-4abc08d17027	Active	2024-09-30	☐
3bfa2157-0171-4357-bb3b-0798a84c1ecc	Invalid	2024-09-30	☐
c49ced9a-59bd-43f5-b0ea-dbf6a6c7a648	Invalid	2024-09-30	☒

New Key

Deactivate

Manage Key

If not select any key and try to deactivate, the page shows an alert.

localhost:8080/APIDeveloperPortal/dashboard.html

localhost:8080 says
Please select a key.

OK

Key	Status	Creation Date	Select
bc533ca6-1713-4e06-9993-b68e4d72a179	Active	2024-09-30	☐
22b06a94-20dd-4666-84cd-5aa676930da3	Active	2024-09-30	☐
a5d4f40c-4a63-4ab6-8d4a-4cad4937bacc	Active	2024-09-30	☐
7ec9ab61-29d3-4253-8ac4-203be0b23347	Invalid	2024-09-30	☐
28302d4d-5d98-4c07-8f52-1b61ac956b27	Active	2024-09-30	☐
029522e2-8f89-4705-8784-c1beea7a0989	Active	2024-09-30	☐
db714111-4d98-4b84-bb10-cbc460e72b7b	Active	2024-09-30	☐
36ea76d1-2fff-44c4-af95-a212a5499fb9	Active	2024-09-30	☐
696d8102-adc1-438c-b99b-4abc08d17027	Active	2024-09-30	☐
3bfa2157-0171-4357-bb3b-0798a84c1ecc	Invalid	2024-09-30	☐
c49ced9a-59bd-43f5-b0ea-dbf6a6c7a648	Invalid	2024-09-30	☐

New Key

Deactivate

Manage Key

Select a key and click “Manage Key” button, will navigate to key management page.

Key	Status	Creation Date	Select
bc533ca6-1713-4e06-9993-b68e4d72a179	Active	2024-09-30	☐
22b06a94-20dd-4666-84cd-5aa676930da3	Active	2024-09-30	☐
a5d4f40c-4a63-4ab6-8d4a-4cad4937bacc	Active	2024-09-30	☐
7ec9ab61-29d3-4253-8ac4-203be0b23347	Invalid	2024-09-30	☐
28302d4d-5d98-4c07-8f52-1b61ac956b27	Active	2024-09-30	☐
029522e2-8f89-4705-8784-c1beea7a0989	Active	2024-09-30	☐
db714111-4d98-4b84-bb10-cbc460e72b7b	Active	2024-09-30	☐
36ea76d1-2fff-44c4-af95-a212a5499fb9	Active	2024-09-30	☐
696d8102-adc1-438c-b99b-4abc08d17027	Active	2024-09-30	☒
3bfa2157-0171-4357-bb3b-0798a84c1ecc	Invalid	2024-09-30	☐
c49ced9a-59bd-43f5-b0ea-dbfaa6c7a648	Invalid	2024-09-30	☐

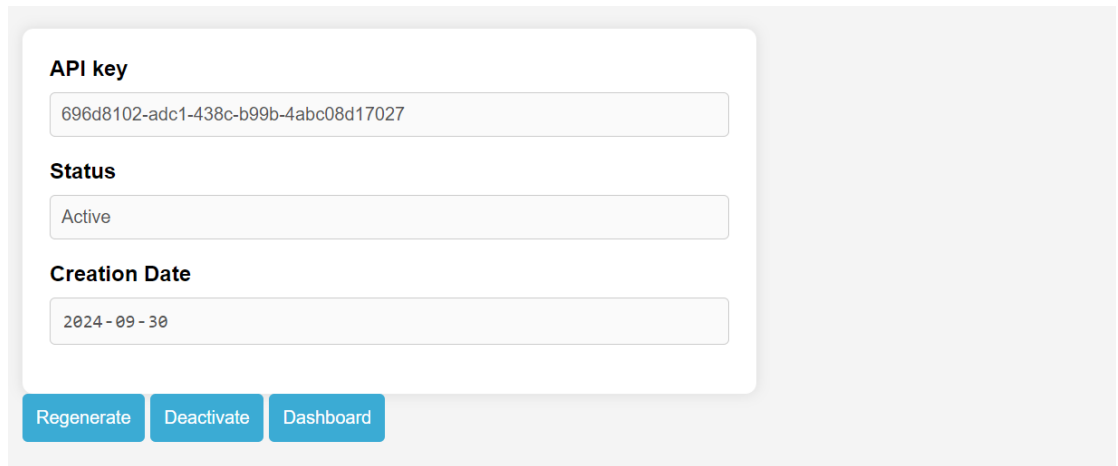
New Key

Deactivate

Manage Key

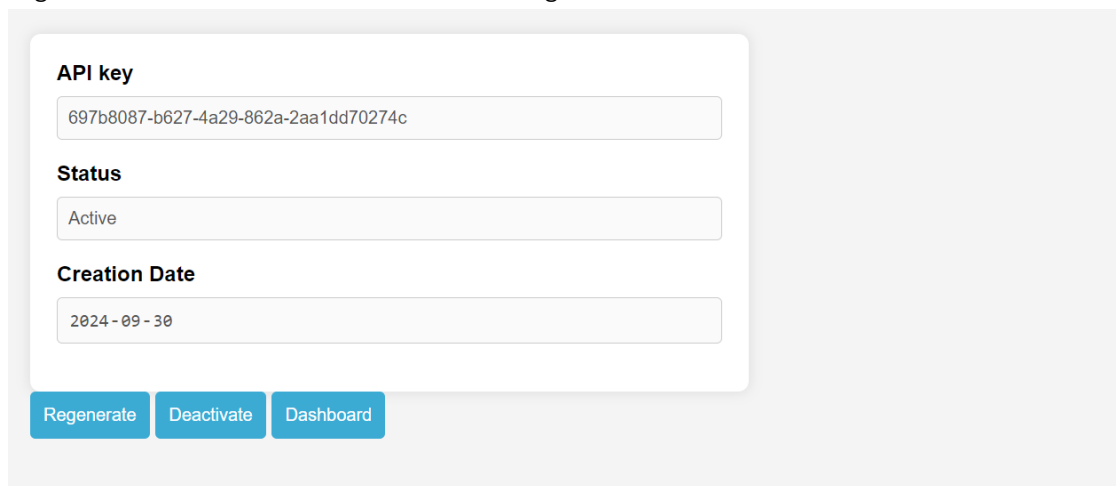
Key management page

When opened, the page displays the information of selected key.



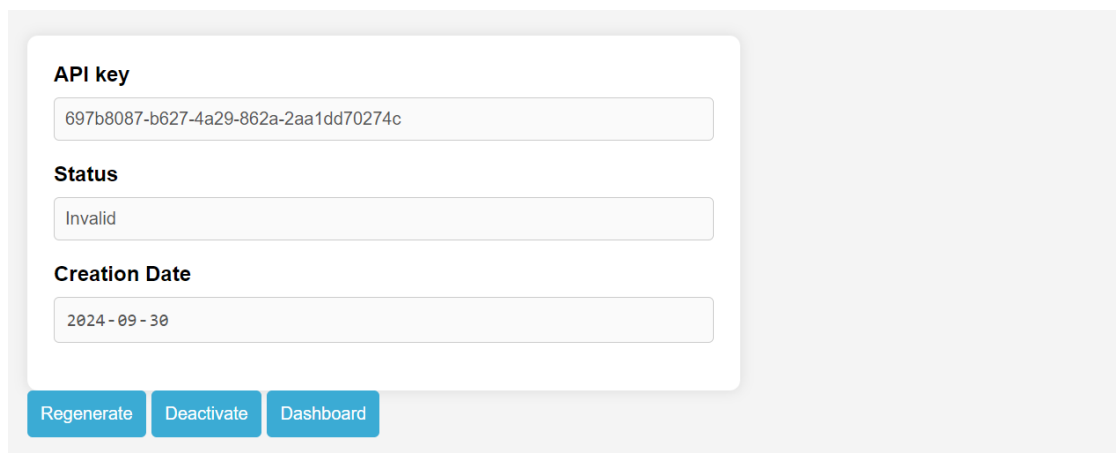
The screenshot shows a white card on a light gray background. The card has three sections: 'API key' with a text input containing '696d8102-adc1-438c-b99b-4abc08d17027', 'Status' with a text input containing 'Active', and 'Creation Date' with a text input containing '2024-09-30'. Below these sections are three blue buttons: 'Regenerate', 'Deactivate', and 'Dashboard'.

Click “Regenerate” button, the key’s value will refresh, and creation date will change to the current day. If the key’s status is invalid, it will be active again after regeneration. While the id remains unchanged.



The screenshot shows the same white card as before, but the 'API key' text input now contains '697b8087-b627-4a29-862a-2aa1dd70274c'. The 'Status' and 'Creation Date' text inputs remain 'Active' and '2024-09-30' respectively. The three blue buttons ('Regenerate', 'Deactivate', 'Dashboard') are still present at the bottom.

Click “Deactivate” button will set the status to invalid.



The screenshot shows the same white card, but the 'Status' text input now contains 'Invalid'. The 'API key' text input remains '697b8087-b627-4a29-862a-2aa1dd70274c' and the 'Creation Date' text input remains '2024-09-30'. The three blue buttons ('Regenerate', 'Deactivate', 'Dashboard') are still present at the bottom.

Click “Dashboard” will navigate to the dashboard page.